

FGFR3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR3 gene fusions from the msk_impact_50k_2026 study, covering 152 fusion events. 74/152 fusions (48.7%) were in-frame. 142/152 fusions (93.4%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0194824$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR3.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 152 analyzed fusion events, identifying 15 distinct fusion partner genes. The most frequent partner was TACC3, observed in 131 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 73/74 (98.6%) of in-frame fusions retained the domain, $p=0.0194824$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: FGFR3 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 151 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 677 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -71 aa from the nearest configured domain boundary at 748 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: FGFR3 has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: FGFR3 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TACC3	131	0.8618421052631579	0.20043213737826426		0.8507823563952885		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.6232543134040097	1
TNIP2	3	0.019736842105263157	0.20043213737826426		0.8763197586726998		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26782435825292855	2
CCDC149	2	0.013157894736842105	0.20043213737826426		0.8778280542986425		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2653280155863073	3
JAKMIP1	2	0.013157894736842105	0.20043213737826426		0.8778280542986425		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2653280155863073	4
LPCAT2	2	0.013157894736842105	0.20043213737826426		0.8770739064856712		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26516641248352774	5
PTBP1	1	0.006578947368421052	0.20043213737826426		0.8778280542986425		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2625084667141268	6
CTBP1	1	0.006578947368421052	0.20043213737826426		0.8763197586726998		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26218526050856766	7
MAEA	1	0.006578947368421052	0.20043213737826426		0.8506787330316742		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2566907550140622	8
ROBO1	1	0.006578947368421052	0.20043213737826426		0.8159879336349924		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24925701228620178	9
NSD2	3	0.019736842105263157	0.20043213737826426		0.6561085972850679		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22063625224129316	10
RBM20	1	0.006578947368421052	0.20043213737826426		0.2413273001508296		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1261154479681669	11
BABAM2	1	0.006578947368421052	0.20043213737826426		0.20965309200603321		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11932811765142481	12
SLIT2	1	0.006578947368421052	0.20043213737826426				['domain_retention', 'recurrence']	0.0946940337365316	13
GRK4	1	0.006578947368421052	0.20043213737826426		0.0015082956259426794		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07472566128426256	14
DGKQ	1	0.006578947368421052	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07440245507870341	15

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
14	1	141	0	9		1.0	-0.0
15	2	140	0	9		1.0	-0.0
42	2	140	1	8	0.11428571428571428	0.1694208631494733	0.7710331099865381
127	3	139	1	8	0.17266187050359713	0.21992905390389725	0.6577173940286984
153	4	138	1	8	0.2318840579710145	0.26768849958325047	0.5723702865449334
174	4	138	2	7	0.10144927536231885	0.04197605082466328	1.3769984229650098
188	4	138	3	6	0.057971014492753624	0.004616748961172982	2.33566373986112
311	4	138	4	5	0.036231884057971016	0.00037923555536042514	3.421090951996021
376	4	138	5	4	0.02318840579710145	2.3640024802728035e-05	4.626352072135864
648	4	138	6	3	0.014492753623188406	1.1022799082261042e-06	5.957708108603845
653	4	138	7	2	0.008281573498964804	3.6665492410410534e-08	7.4357424781390336
677	4	138	8	1	0.0036231884057971015	7.825580002902196e-10	9.106483464295888
753	5	137	8	1	0.004562043795620438	2.021680915172072e-09	8.694287388663584
758	52	90	8	1	0.07222222222222222	0.002807318578004277	2.5517083002909002

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
759	88	54	8	1	0.2037037037037037	0.15623087834924132	0.806233125571681
760	89	53	8	1	0.2099056603773585	0.15840511374693833	0.8002308023157874
762	91	51	8	1	0.22303921568627452	0.16500164368454323	0.7825117294856598
763	93	49	8	1	0.2372448979591837	0.27269821801700755	0.5643177000044979
764	95	47	8	1	0.2526595744680851	0.2731204045238703	0.5636458529869772
766	96	46	8	1	0.2608695652173913	0.27473748927465147	0.5610820749288234
767	97	45	8	1	0.26944444444444443	0.27730934214956404	0.5570354994244366
768	100	42	8	1	0.2976190476190476	0.4468103647595553	0.34987676101374454
769	101	41	8	1	0.3079268292682927	0.4455794616522395	0.3510748355557522
770	105	37	8	1	0.3547297297297297	0.45054685984922543	0.34626003284121815
771	108	34	8	1	0.39705882352941174	0.6854647233823904	0.16401489074257258
772	111	31	8	1	0.4475806451612903	0.6849966189342125	0.16431157213069608
773	113	29	8	1	0.4870689655172414	0.688608956187972	0.16202733307235542
776	115	27	8	1	0.5324074074074074	1.0	-0.0
777	119	23	8	1	0.6467391304347826	1.0	-0.0
778	120	22	8	1	0.6818181818181818	1.0	-0.0
779	121	21	8	1	0.7202380952380952	1.0	-0.0
780	122	20	8	1	0.7625	1.0	-0.0
781	125	17	8	1	0.9191176470588235	1.0	-0.0
782	127	15	8	1	1.0583333333333333	1.0	-0.0
785	128	14	8	1	1.1428571428571428	1.0	-0.0
786	130	12	8	1	1.3541666666666667	0.5655253555991285	0.24754791850170255
787	131	11	9	0	0.0	1.0	-0.0
788	133	9	9	0	0.0	1.0	-0.0
789	134	8	9	0	0.0	1.0	-0.0
793	138	4	9	0	0.0	1.0	-0.0
797	139	3	9	0	0.0	1.0	-0.0
799	140	2	9	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

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73	69
1	8

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein tyrosine and serine/threonine kinase	151	142	3	6	0.22663457681508226

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
BABAM2	1
CCDC149	2
CTBP1	1
DGKQ	1
GRK4	1
JAKMIP1	2

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event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0022387-T02-IM6-48	P-0022387-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+46:FGFR3_c.1942-224:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-70:FGFR3_c.1645-159:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	
EVT-P-0044466-T02-IM6-49	P-0044466-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808773	17	759	True	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-70:FGFR3_c.1645-159:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	
EVT-P-0055884-T01-IM6-50	P-0055884-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808886	17	773	False	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	
EVT-P-0000941-T02-IM6-51	P-0000941-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808894	17	776	False	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	
EVT-P-0056846-T01-IM6-52	P-0056846-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808698	16	758	True	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+46:FGFR3_c.1942-224:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-70:FGFR3_c.1645-159:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	
EVT-P-0053694-T03-IM6-53	P-0053694-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	
EVT-P-0053694-T02-IM6-54	P-0053694-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808911	17	781	False	retained	1.0	False						FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0011503-T01-IM5-64	P-0011503-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808967	17	800	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2399:FGFR3_c.1836+	NA	75:TACC3dup
EVT-P-0051755-T01-IM6-65	P-0051755-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808806	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-37:FGFR3_c.1942-333:TACC3dup	NA	
EVT-P-0010534-T01-IM5-66	P-0010534-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808859	17	764	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 10-16): c.2363:FGFR3_c.1784:TACC3dup	NA	
EVT-P-0010096-T03-IM6-67	P-0010096-T03-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808925	17	786	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2357:FGFR3_c.1941+	NA	378:TACC3dup
EVT-P-0060879-T02-IM7-68	P-0060879-T02-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808945	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1942-241:TACC3dup	NA	
EVT-P-0062326-T01-IM7-69	P-0062326-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808913	17	782	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2345:FGFR3_c.1386-12:TACC3dup	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	is_maintained	deleted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0045382-T01-IM6-91	P-0045382-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808903	17	779	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2335:FGFR3_c.1942-101:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2276:FGFR3_c.1942-892:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-62:FGFR3_c.1837-81:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA	
EVT-P-0047573-T01-IM6-92	P-0047573-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808844	17	759	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2276:FGFR3_c.1942-892:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-62:FGFR3_c.1837-81:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA	
EVT-P-0064650-T01-IM7-93	P-0064650-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808781	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-62:FGFR3_c.1837-81:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA	
EVT-P-0065466-T01-IM7-94	P-0065466-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808795	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA	
EVT-P-0005361-T01-IM5-96	P-0005361-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1803212	4	188	False	lost	0.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA	
EVT-P-0005128-T01-IM5-97	P-0005128-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808814	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0032266-T01-IM6-98	P-0032266-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808745	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+84:FGFR3_c.1645-165:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+45:FGFR3_c.1942-654:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3		
EVT-P-0016238-T03-IM7-99	P-0016238-T03-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+45:FGFR3_c.1942-654:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3		
EVT-P-0047604-T02-IM6-100	P-0047604-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808692	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_c.1941+484:TACC3		
EVT-P-0066401-T01-IM7-101	P-0066401-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808930	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2362:FGFR3_c.1942-935:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2313:FGFR3_c.1837-88:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: (FGFR3 exon 17 fused with TACC3 exon 9) : c.2313:FGFR3_c.1837-88:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) Protein fusion: in frame (FGFR3-TACC3)		
EVT-P-0004694-T01-IM5-102	P-0004694-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1808881	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2313:FGFR3_c.1837-88:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) Protein fusion: in frame (FGFR3-TACC3)		
EVT-P-0003853-T01-IM5-103	P-0003853-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808680	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) Protein fusion: in frame (FGFR3-TACC3)		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	breakpoint	main_ref	donor	acceptor	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0038318-T01-IM6-110	P-0038318-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808756	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-87:FGFR3_c.1462-57:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2301:FGFR3_c.1941+974:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		
EVT-P-0038414-T01-IM6-111	P-0038414-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808869	17	767	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2301:FGFR3_c.1941+974:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		
EVT-P-0039109-T01-IM6-112	P-0039109-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		
EVT-P-0039109-T02-IM6-113	P-0039109-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808760	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		
EVT-P-0039116-T01-IM6-114	P-0039116-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808698	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		
EVT-P-0039138-T02-IM6-115	P-0039138-T02-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808645	16	753	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		
EVT-P-0039166-T01-IM6-116	P-0039166-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808883	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	inframe	deleted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0067303-T01-IM7-117	P-0067303-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808812	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-31:FGFR3_c.1941+863:TACC3dup		
EVT-P-0039655-T01-IM6-118	P-0039655-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808887	17	773	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2319:FGFR3_c.1836+86:TACC3dup		
EVT-P-0003279-T02-IM6-119	P-0003279-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.2274+47:FGFR3_c.1941+850:TACC3		
EVT-P-0040543-T01-IM6-120	P-0040543-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808696	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+35:FGFR3_c.1836+39:TACC3d		
EVT-P-0067315-T01-IM7-121	P-0067315-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808877	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2309:FGFR3_c.1941+162:TACC3dup		
EVT-P-0067602-T01-IM7-122	P-0067602-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808775	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-68:FGFR3_c.1836+111:TACC3dup		
EVT-P-0047905-T01-IM6-123	P-0047905-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808802	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-41:FGFR3_c.1837-96:TACC3dup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_disrupted	description	annotation	effect_on_fra
EVT-P-0054391-T01-IM6-132	P-0054391-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808789	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-54:FGFR3_c.1941+447: TACC3dup		TACC3dup
EVT-P-0006429-T01-IM5-133	P-0006429-T01-IM5		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808801	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TNIP2 (NM_024309) rearrangement: c.2275-42:FGFR3_c.147: TNIP2inv		
EVT-P-0006429-T02-IM6-134	P-0006429-T02-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808753	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-90:FGFR3_c.195: TNIP2inv		
EVT-P-0050971-T01-IM6-135	P-0050971-T01-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			Fusion: mid-exon {FGFR3:TNIP2} FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-16:FGFR3_c.252: TNIP2inv		
EVT-P-0012257-T01-IM5-136	P-0012257-T01-IM5		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808759	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			Fusion: mid-exon {FGFR3:TNIP2} FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16_3;q12.2)(chr4:g.1808759::chr		
EVT-P-0012257-T03-IM6-137	P-0012257-T03-IM6		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			Fusion: in frame {FGFR3:LPCAT2} FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16_3;q12.2)(chr4:g.1808708::chr		
EVT-P-0024146-T01-IM6-139	P-0024146-T01-IM6		NSD2-FGFR3	NSD2	in-frame	three_prime	1801414	3	127	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			Fusion: in frame {FGFR3:WHSC1} WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.761-178:WHSC1_c.380-60:FGFR3dup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein_pos	protein_pos	protein_pos	main_retained	donor_inframe	donor_inframe	disrupted_domain	disrupted_domain	disrupted_domain	effect_on_frame
EVT-P-0031155-T01-IM6-153	P-0031155-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808715	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+54:FGFR3_c.1644+125:TACC3	NA	
EVT-P-0026783-T01-IM6-154	P-0026783-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808848	17	760	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2280:FGFR3_c.1942-78:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0024954-T02-IM6-155	P-0024954-T02-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808755	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0024954-T01-IM6-156	P-0024954-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808755	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3_c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	main_ref	donor_frame	is_maintained	donor_domain	disrupted_domain	description	annotation	effect_on_frame
EVT-P-0002351-T01-IM3-157	P-0002351-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808871	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) and TACC3 (NM_006342) duplication (71843bp): FGFR3:c.2303_TACC3:c.1620dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 6-16): c.2291:FGFR3_c.1461+69:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 4-16): c.2315:FGFR3_c.927:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c.1941+873:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0024346-T01-IM6-158	P-0024346-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808859	17	764	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 6-16): c.2291:FGFR3_c.1461+69:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 4-16): c.2315:FGFR3_c.927:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c.1941+873:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0033514-T01-IM6-159	P-0033514-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808883	17	772	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 4-16): c.2315:FGFR3_c.927:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c.1941+873:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0023261-T01-IM6-160	P-0023261-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808927	17	787	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c.1941+873:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0022569-T01-IM6-161	P-0022569-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808705	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGFR3_c.1645-194:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	deleted	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0017056-T01-IM6-167	P-0017056-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808671	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 11-16): c.2274+10:FGFR3_c.1941+881:TACC3dup	NA	
EVT-P-0016989-T01-IM6-168	P-0016989-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808718	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 8-16): c.2274+57:FGFR3_c.1645-139:TACC3dup	NA	
EVT-P-0001453-T01-IM3-169	P-0001453-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase					NA NA	NA	
EVT-P-0001453-T01-IM3-170	P-0001453-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase					NA NA	NA	
EVT-P-0016238-T01-IM6-171	P-0016238-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 11): c.2274+45:FGFR3_c.1942-654:TACC3inv	NA	
EVT-P-0015863-T01-IM6-173	P-0015863-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808880	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2312:FGFR3_c.1644+147:TACC3dup	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_neg	main_retention	dominant_retention	protein_pos	protein_neg	disrupted_domains	domains_annotation	effect_on_fra	
EVT-P-0012373-T01-IM5-179	P-0012373-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808730	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16): c.2274+69:FGFR3_c.1837-70:TACC3dup Protein fusion: in frame (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+69:FGFR3_c.1837-70:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 - NA (NM_000142) TACC3 (NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon3) : c.2334:FGFR3_c.274:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3): c.2310:FGFR3_c.1645-1546:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0034666-T01-IM6-180	P-0034666-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808721	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+60:FGFR3_c.1941+226:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 - NA (NM_000142) TACC3 (NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon3) : c.2334:FGFR3_c.274:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3): c.2310:FGFR3_c.1645-1546:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0036305-T01-IM6-181	P-0036305-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808902	17	778	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3): c.2310:FGFR3_c.1645-1546:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0008649-T01-IM5-182	P-0008649-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808878	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3): c.2310:FGFR3_c.1645-1546:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0029600-T01-IM6-183	P-0029600-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808697	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3_c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_positives	in_frame	main_annotation	donor_inframe	retained_domains	disrupted_domains	domains_annotation	station_site	effect_on_fra
EVT-P-0007082-T03-IM6-184	P-0007082-T03-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808872	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR3(NM_000142) - TACC3(NM_006342) Fusion (FGFR3 exon 18 fused to TACC3 exon 8) : c.2304:FGFR3_c.1644+	118:TACC3dup	
EVT-P-0007082-T02-IM6-185	P-0007082-T02-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808872	17	768	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase		Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 8-16 of TACC3): c.2304:FGFR3_1644+	118:TACC3dup	
EVT-P-0005966-T01-IM5-186	P-0005966-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808853	17	762	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase		Fusion: mid-exon {FGFR3:TACC3} FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) : c.2285:FGFR3_c.1941+	423:TACC3dup	
EVT-P-0005862-T01-IM5-187	P-0005862-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		Fusion: mid-exon (FGFR3-TACC3) FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion (exons 1-17 of FGFR3 fused to exons 14-16 of TACC3): c.2274+46:FGFR3_c.2223+896:TACC3		
EVT-P-0004902-T01-IM5-189	P-0004902-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808906	17	780	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase		fusion: in frame (FGFR3-TACC3) FGFR3 NA (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2338:FGFR3_c.1941+	300TACC3dup	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	clonally_deleted	inframe	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fusion
EVT-P-0003677-T01-IM5-190	P-0003677-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808931	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				c.2363:FGFR3_c.1784:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3) NA	NA	
EVT-P-0003097-T01-IM5-191	P-0003097-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				Protein fusion: in frame (FGFR3-TACC3) FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+86:FGFR3_c.162+46:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2311:FGFR3_c.1644+71:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} NA	NA	
EVT-P-0003696-T01-IM6-192	P-0003696-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808747	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+86:FGFR3_c.162+46:TACC3dup Protein Fusion: in frame {FGFR3:TACC3} FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2311:FGFR3_c.1644+71:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} NA	NA	
EVT-P-0016735-T01-IM6-193	P-0016735-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808879	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2311:FGFR3_c.1644+71:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3} NA	NA	
EVT-P-0002862-T01-IM3-194	P-0002862-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808873	17	769	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16): c.2275-42:FGFR3_c.1836+11:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	
EVT-P-0024463-T01-IM6-195	P-0024463-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808801	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16): c.2275-42:FGFR3_c.1836+11:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA	

3 Figures

Protein tyrosine and serine/threonine kinase (472-748)

● fully retained ● truncated ● fully lost ○ reference discrepancy

Figure 1: domain retention outliers

partner block → breakpoint → retained FGFR3 portion (domain-colored, exon ticks)



Showing the top 28 of 52 partner/breakpoint groups by recurrence: see results.tsv for the complete list.

Figure 2: fusion schematic

FGFR3 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

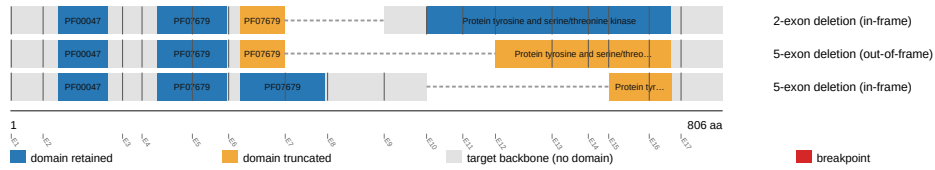


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison